

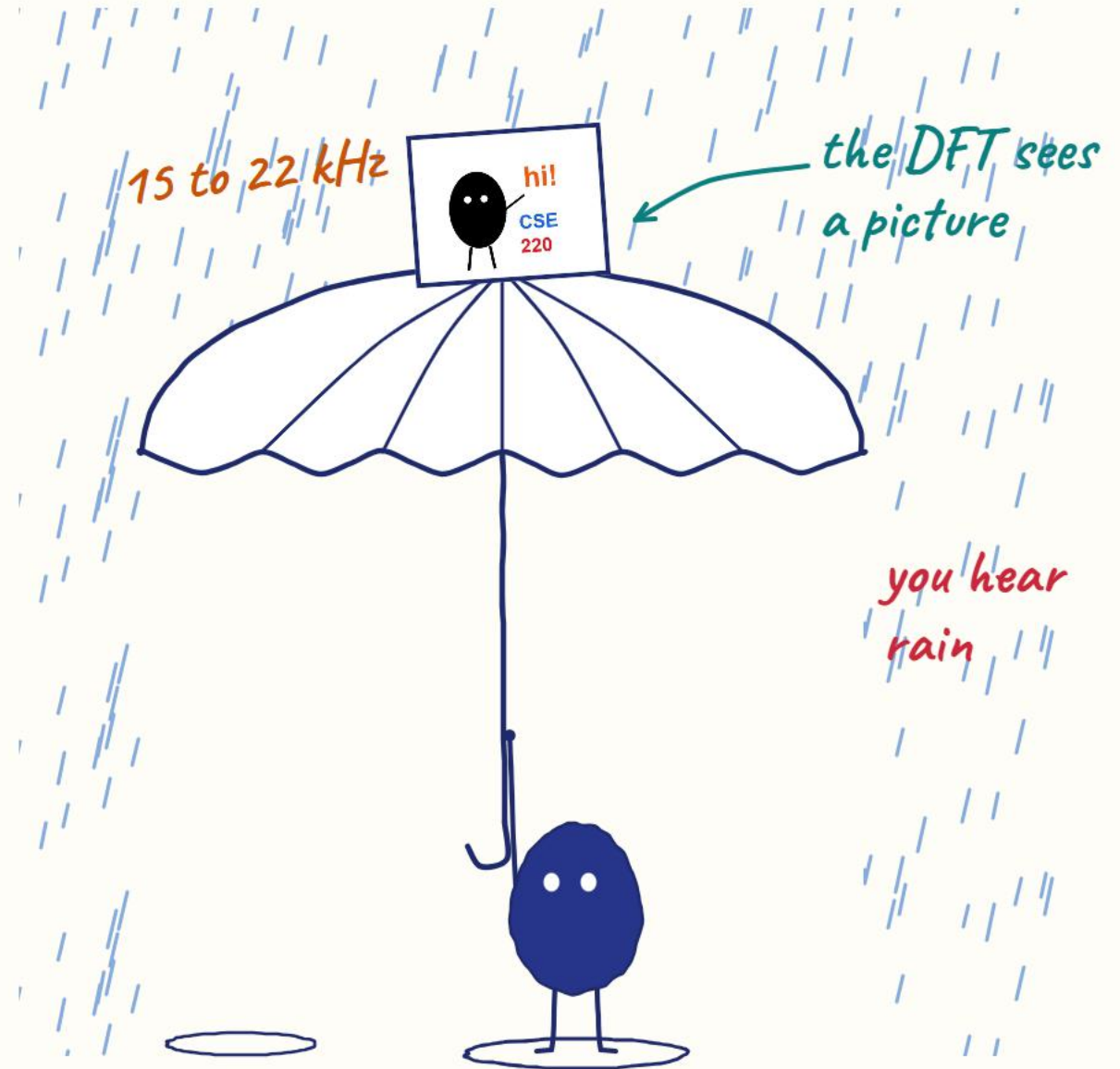
HiddenHz

A picture hidden in the rain

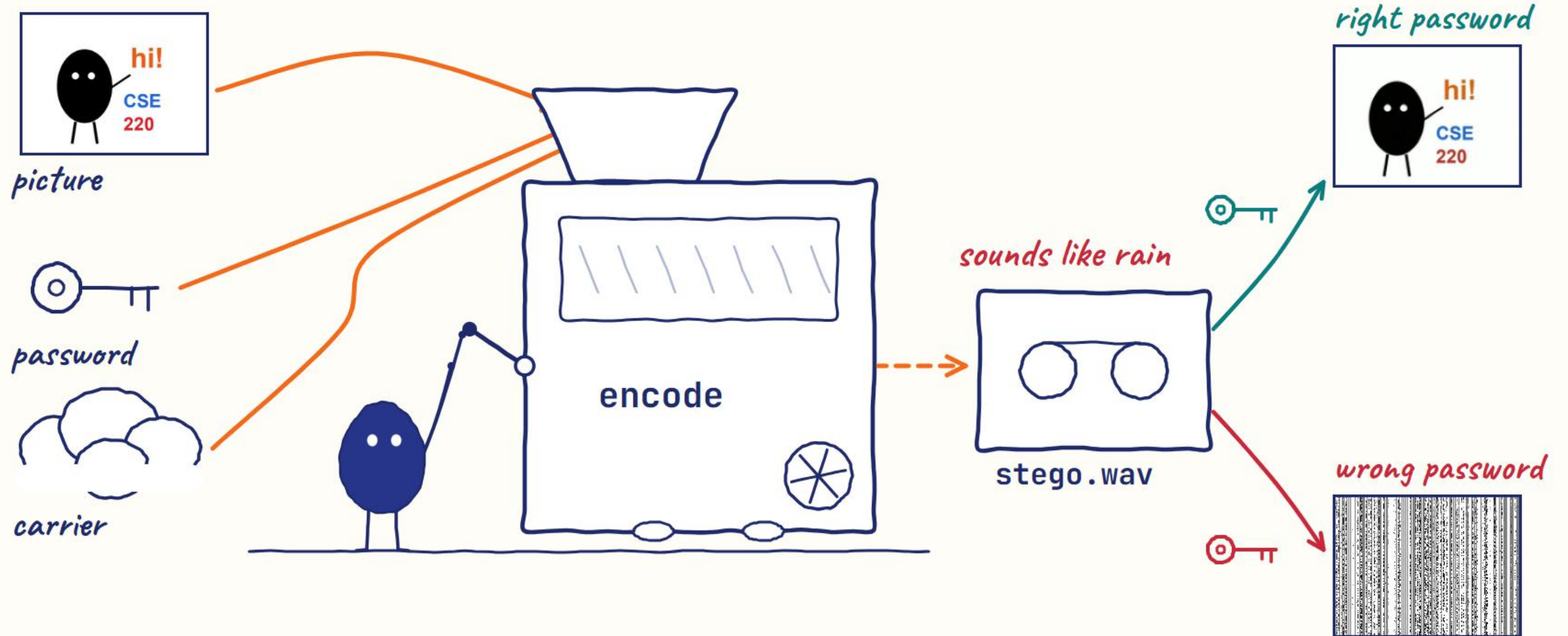
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CSE 220 · Signals and Linear Systems

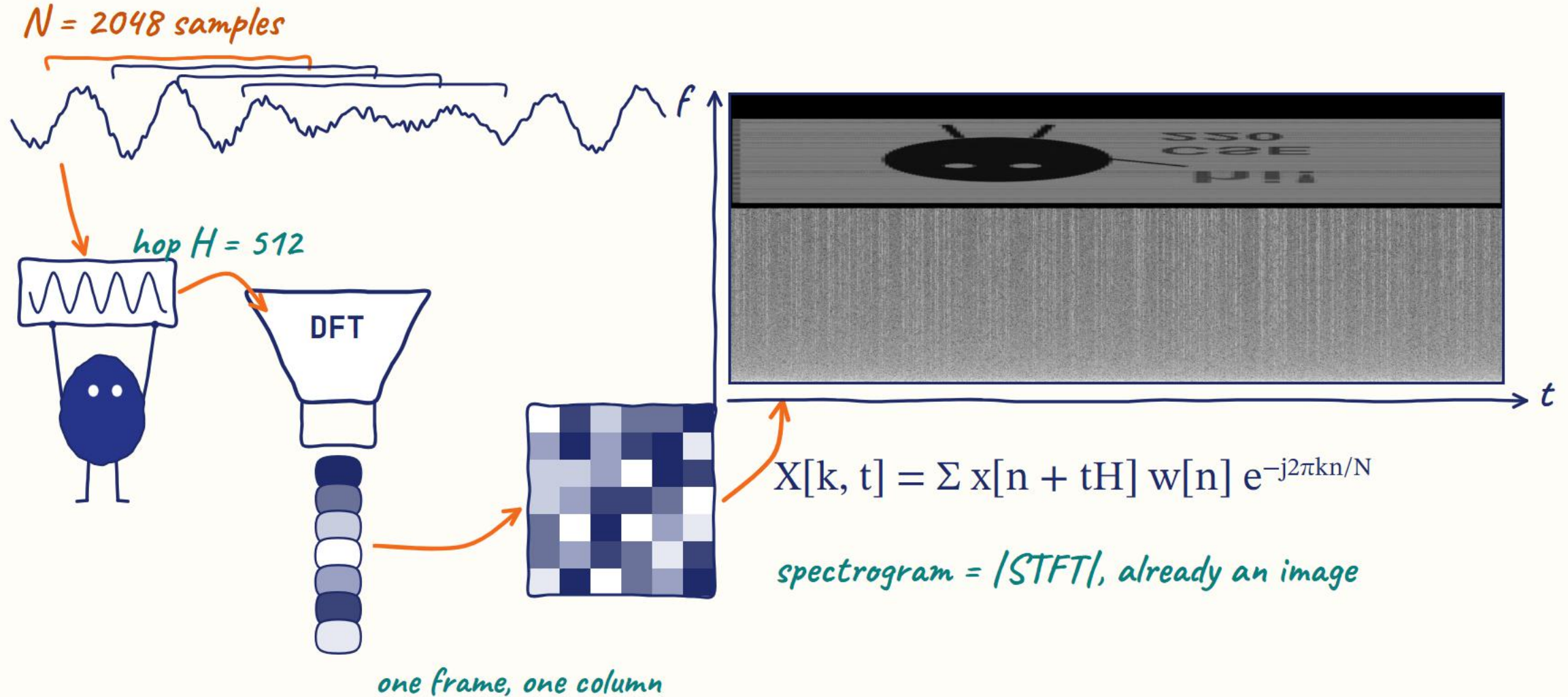


How it works



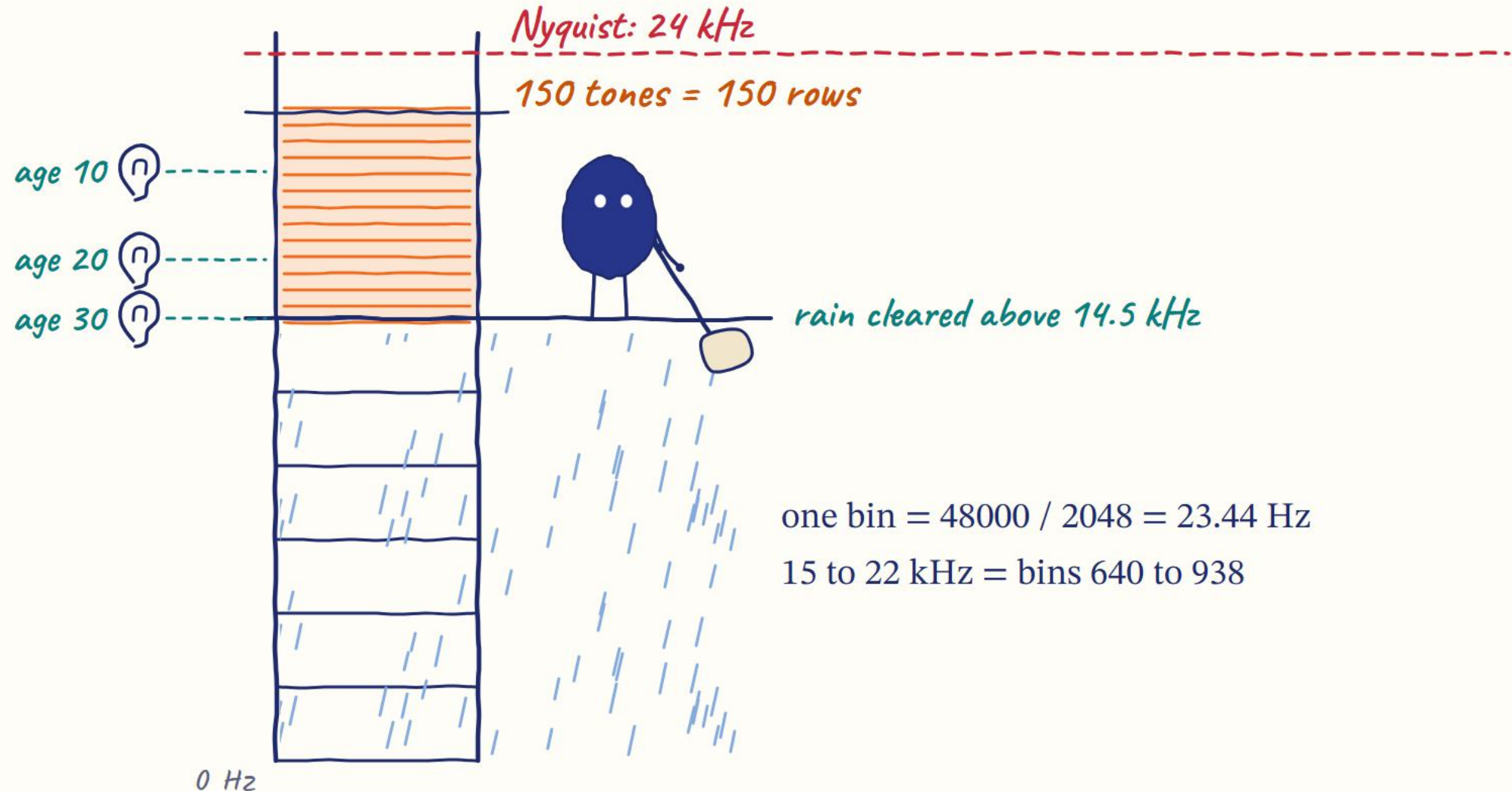
THEORY

A spectrogram is an image



THEORY

Where the picture lives: 15 to 22 kHz



one bin = $48000 / 2048 = 23.44$ Hz

15 to 22 kHz = bins 640 to 938

THEORY

Hann window: one tone, three bins

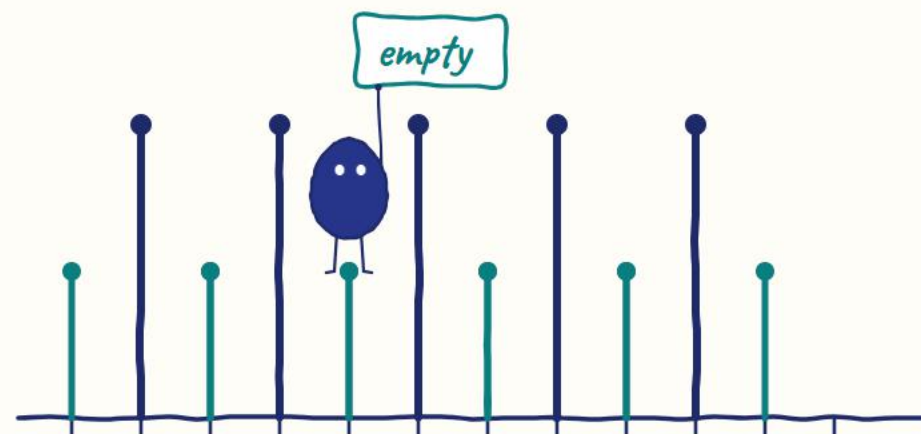
$$w[n] = 0.5 - 0.5 \cos(2\pi n / N)$$

tapers the edges: a tone fills exactly 3 bins



a chopped tone leaks everywhere

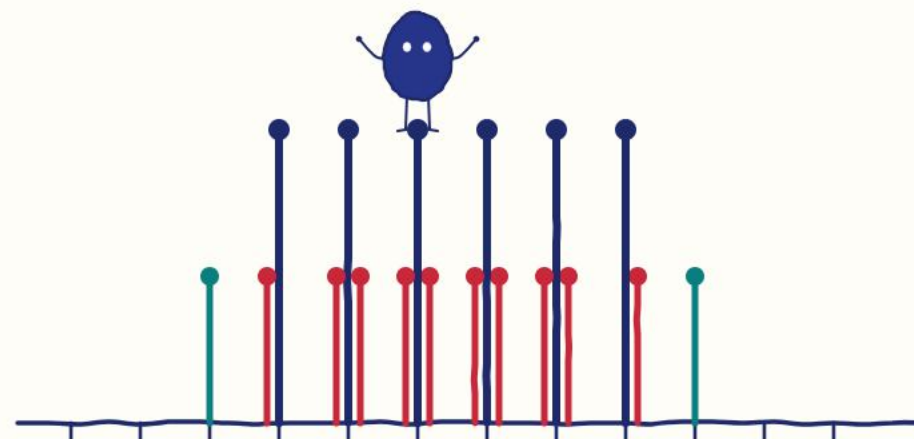
spacing 2 (ours)



quality 0.98

HiddenHz

spacing 1

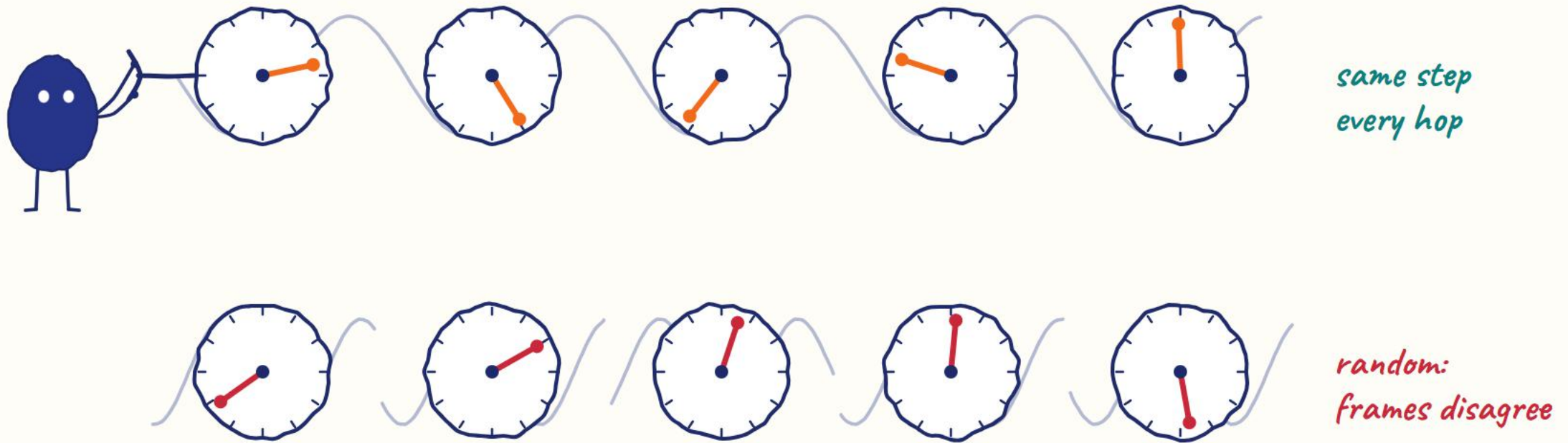


quality 0.59

THEORY

Back to sound: the inverse STFT

$$\varphi[k, t] = \varphi_0[k] + 2\pi k H t / N$$

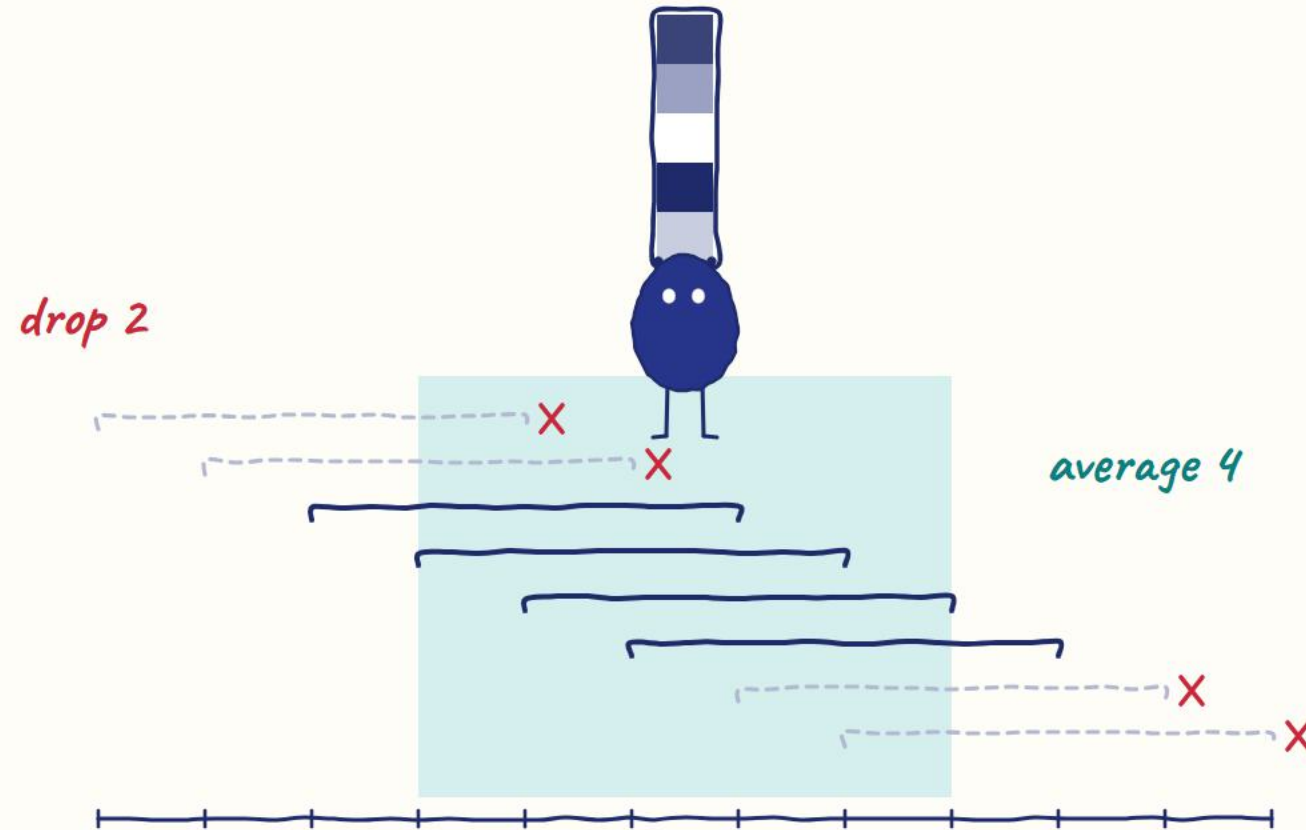


inverse DFT each column, overlap-add, 4 blank frames at the ends

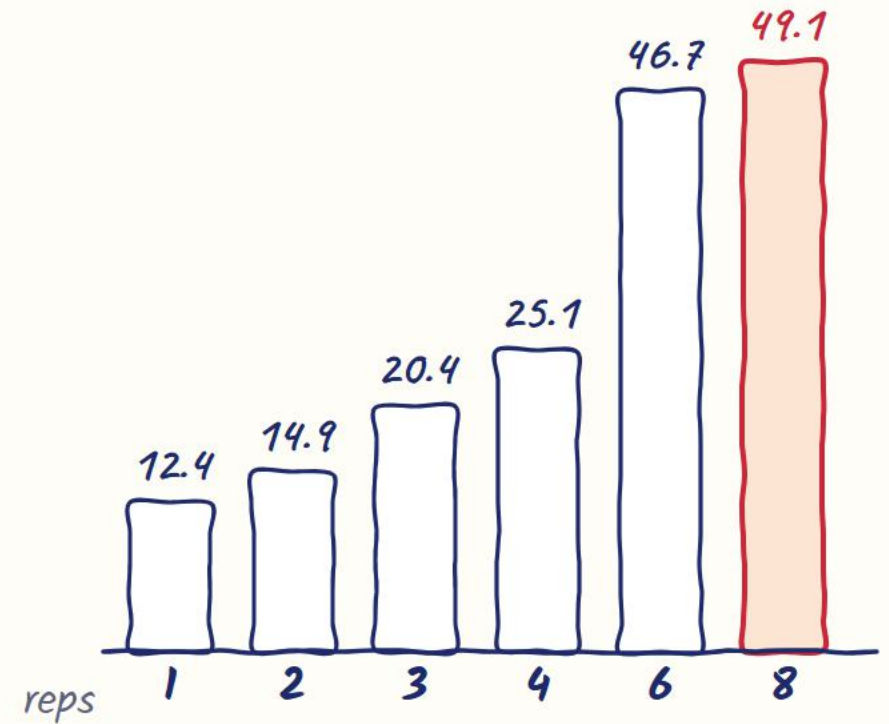
THEORY

Each column lasts eight frames

a frame spans 4 hops, so a column must last longer



PSNR (dB)



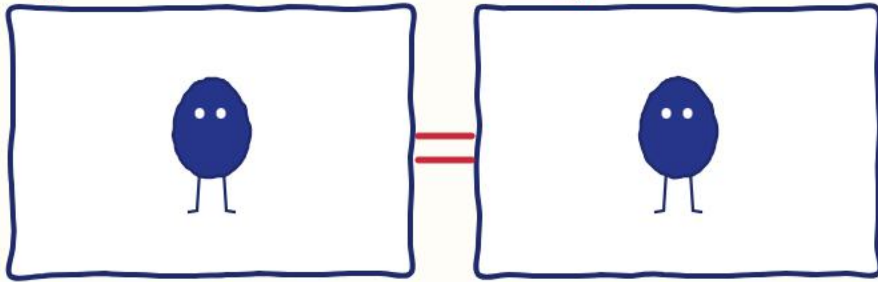
SECURITY

What the password does

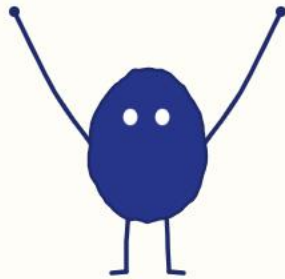
random phase?

right

wrong

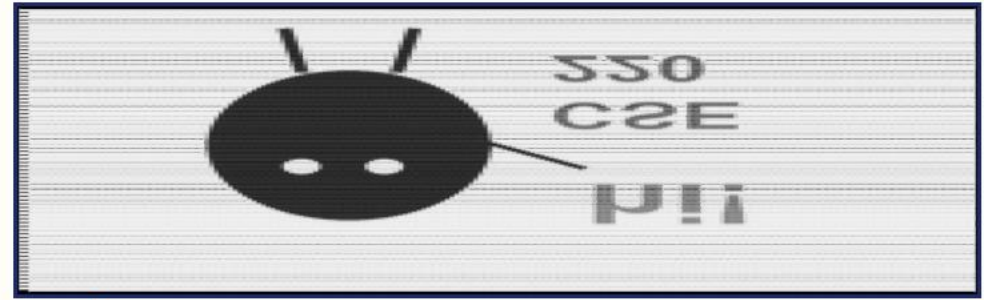


$$0.914 = 0.914$$



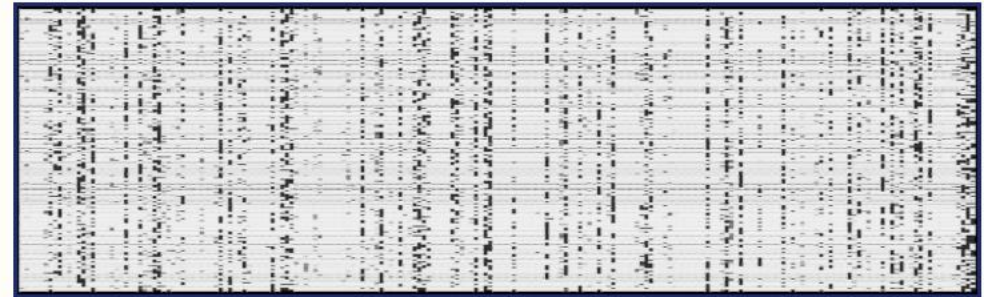
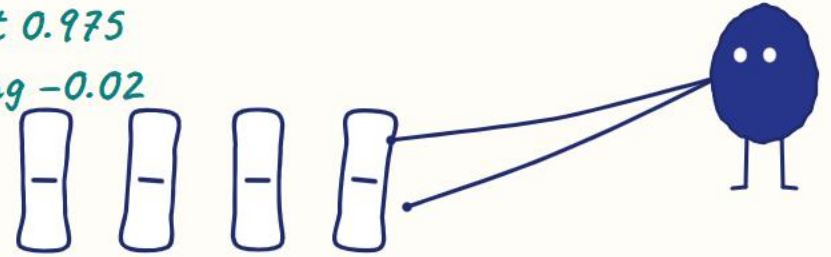
a spectrogram ignores phase

shuffle rows and columns



right 0.975

wrong -0.02



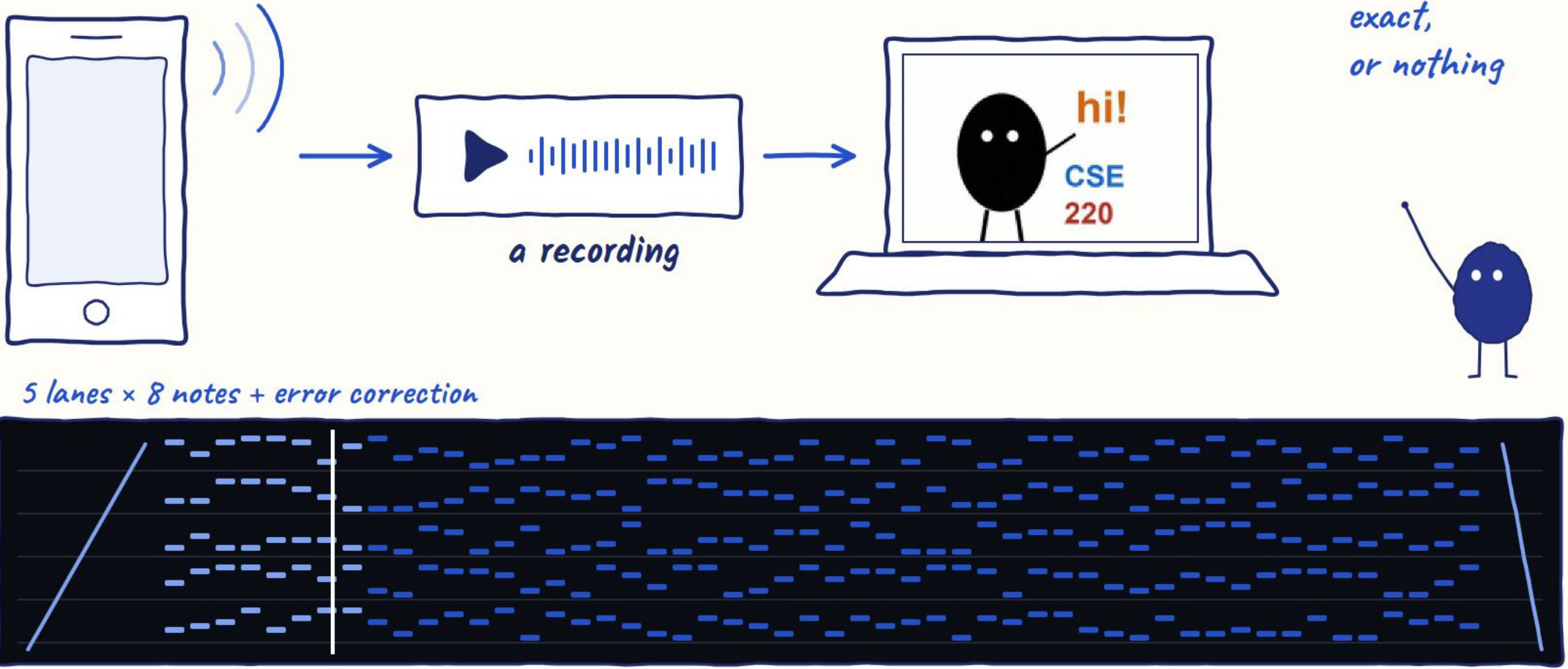
AIR MODE

Air mode: through phones

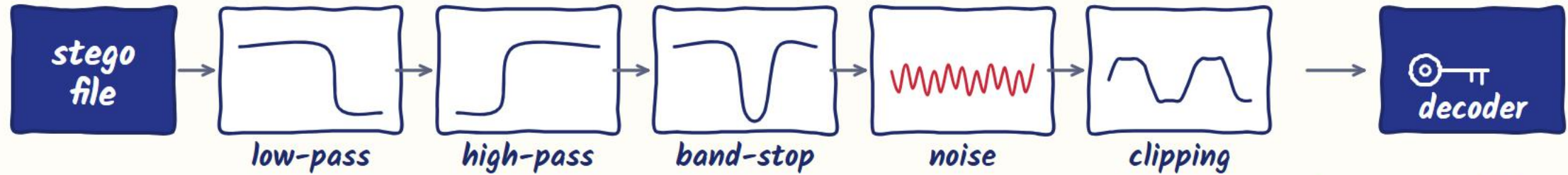
15–22 kHz: gone after a phone

0.75–4.8 kHz: survives

exact,
or nothing



The lab: attack your own file

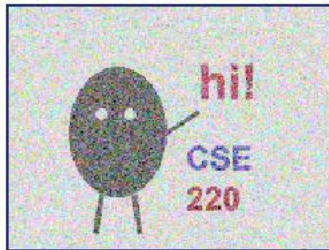


noise sweep



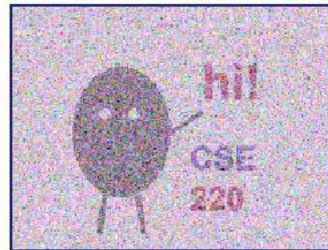
40 dB

survived



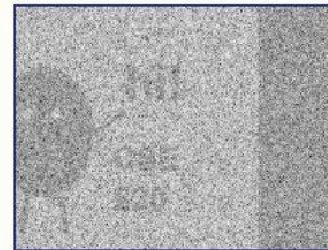
20 dB

survived



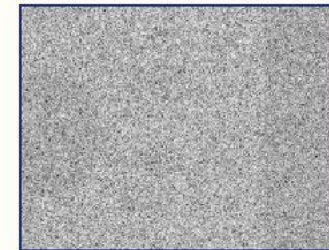
15 dB

damaged



10 dB

damaged



5 dB

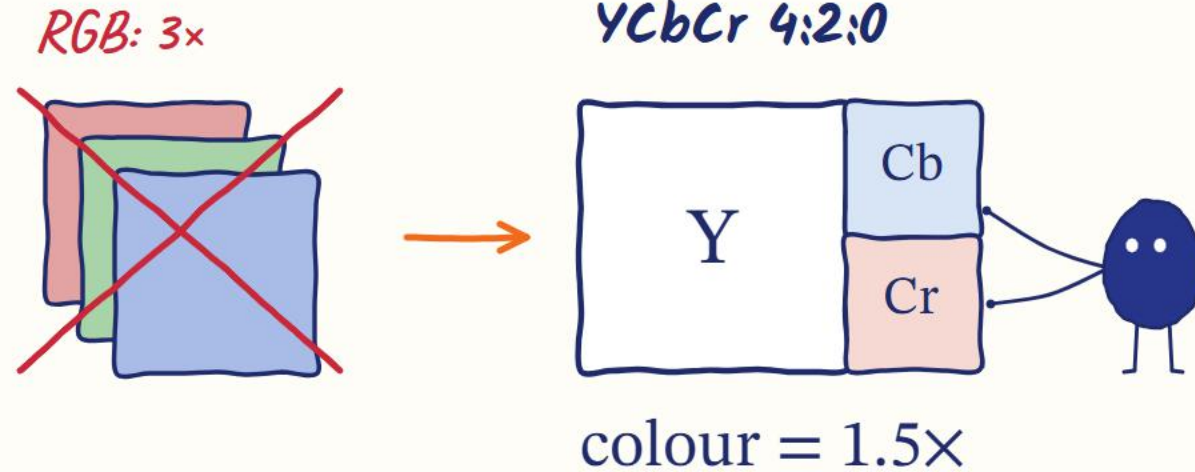
lost

high-pass 10 kHz:
survives

low-pass 18 kHz:
lost



What it costs, where it breaks



150 x 150 picture

13 s grey

19.5 s colour



files: 125 bytes / s



MP3 and AAC
erase the band



anyone can
see it is there

Thank you

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